MiSTer Monitor — setup guide

v2.12.0

● **What is MiSTer Monitor** — What it does, and what you will need. — page 1

1 **Flash the display** — From a browser, over USB. The only time the cable is needed. — page 1

2 **Prepare the display's microSD card** — Copy the ready-made card content and fill in your WiFi. — page 2

3 **Create a free ScreenScraper account** — Free and instant. It is where the artwork comes from. — page 3

4 **Install the server on the MiSTer** — One toggle in Update All or MiSTer Companion — it updates itself from then on — page 3

5 **RetroAchievements (optional)** — Your progress and live unlock popups on the display — page 4

6 **The web interface** — Edit settings and manage the display's card from any browser — page 5

OK **You're done — a quick check** — What a working setup looks like. — page 7

↑ **Updates over WiFi** — From 2.12.0 on, new versions reach the display through the MiSTer — no cable — page 7

.ini **Everything config.ini can do** — Every option, grouped by what you want to achieve — all of it optional — page 8

github.com/chipster6502/MiSTer_monitor

● What is MiSTer Monitor

MiSTer Monitor turns an ESP32-based touchscreen into a companion display for your MiSTer: the moment you load a game, the screen shows its box art, a “Now Playing” panel with related info (year, dev, publisher, genre, players, rating and synopsis) of the active game, your **RetroAchievements** progress, and live system stats. Everything is fetched on demand, and when the MiSTer is switched off the screen dims to a standby clock.

Two pieces work together: a tiny **server** on the MiSTer that watches what you play, and the **firmware** on the display that renders it. You will install both.

You will need

- A **MiSTer** connected to your home network.
- A supported display: **M5Stack Tab5** (5"), **Guition JC8012P4A1C** (10.1"), a **Cheap Yellow Display** (2.8" ILI9341/ST7789, or 3.5" capacitive/resistive), or a **Waveshare ESP32-S3-Touch-LCD-2.8 V2** (2.8", beta).
- A **microSD card** for the display.
- A **USB data cable** and a desktop computer with **Chrome or Edge** (only for the one-time flash).
- **USB power** for the display: a phone charger or a powered USB hub rather than a USB port on the MiSTer, which may not supply enough current for the display's WiFi.
- A free **ScreenScraper** account — created in step 3.
- Optional: a free **RetroAchievements** account — step 5.

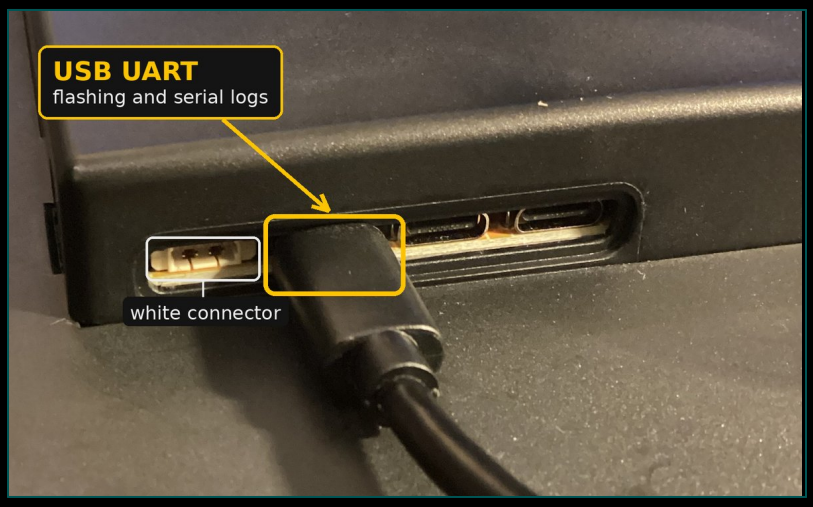
1 Flash the display

- Connect the display to your computer with the USB cable.
- Open the flasher page in **Chrome or Edge** (Firefox and Safari don't support Web Serial):

https://chipster6502.github.io/MiSTer_monitor/flasher/

- Click **Connect** next to your board, pick the serial port in the browser dialog, and wait for the confirmation.
- Not sure which CYD variant you have? The page itself explains how to tell them apart.
- **Waveshare 2.8"** — it has to be the **V2** board, and the page lists it as beta: its touchscreen has not been verified on hardware yet. It has a single USB-C port; if the flasher cannot see it, hold **BOOT** while plugging it in.
- The cable is only needed this once. From 2.12.0 on, new versions reach the display over WiFi — see **Updates over WiFi**, after step 6.

Guition 10.1" — which port, which build



The USB UART port sits right next to the small white connector.

- Flash through the USB-C port marked **USB UART**, the one next to the small white connector. The other two power the board, but the flasher cannot connect through them.
- The most recent units carry a newer ESP32-P4 chip (v3) and need their own build. So far they have a sticker SKU ending in **-V3**, like 10153001-V3 (2637). The flasher page cannot install it yet: download mister_monitor_guition10inch_v3.bin from that page and flash it at address 0x0 from Espressif's browser tool, with the baud rate set to **115200** before connecting. The page has the steps, and an esptool command too.
- Power it from a **5 V / 2 A** supply, such as a phone charger. Some MiSTer USB ports cannot feed the 10.1" panel with WiFi, and the board restarts as the WiFi comes up. It talks to the MiSTer over WiFi, so it does not need to be plugged into it.

2 Prepare the display's microSD card

The repository ships a ready-made card layout under **SD_card_content/**. Pick the subfolder matching your board, copy its contents to the **root** of the microSD card, then open **config.ini** in any text editor and fill in your network — **without quotes** around the values:

```
config.ini — on the display's microSD card
[wifi]
ssid=YOUR_WIFI_NAME
password=YOUR_WIFI_PASSWORD
```

That is all the display needs to find the MiSTer: it discovers it automatically on the network at boot.

This first edit has to happen on a computer, because the display needs the WiFi details before it can be reached over the network. From then on you can edit **config.ini** from a browser instead — see step 6.

NOTE With a single MiSTer, leave **[mister] ip=** blank. With several MiSTers on the network, set it to the address of the one this display should follow — it will never adopt another. Set it too if your router blocks UDP broadcast, which is rare in home networks.

IMPORTANT Guition 10.1" only. Guition ships this board with two different panels. If the screen shows horizontal colour bands instead of the interface, yours is the newer one — batch 2624 onward, the figure in parentheses after the SKU on the back sticker, and every **-V3** unit — and needs panel_rev=v2 in the **[ui]** section. Keep c6_firmware.bin in the root of the card too: if the board's WiFi chip carries firmware too old to connect, the display updates it from that file at boot, by itself. Take it from the same release as the firmware; the display refuses an older one.

TIP Guition: WiFi stopped connecting after you opened the case? The back cover has to come off to reach the card, and the small antenna cable inside snaps off easily. Check that it is still connected.

3 Create a free ScreenScraper account

- Register at **screenscraper.fr** — free, instant, no approval needed.
- Add your username and password to the same **config.ini**:

```
config.ini — same file, next section
[screenscraper]
ss_user=YOUR_SS_USERNAME
ss_pass=YOUR_SS_PASSWORD
```

The app brings its own developer credentials, so a plain member account is enough. Artwork is downloaded the first time each game appears and cached on the card, so it costs quota once and loads instantly forever after.

4 Install the server on the MiSTer

One toggle in Update All or MiSTer Companion — it updates itself from then on

Pick whichever updater you already use. Both install the same files and keep them current from then on.

- **Update All** — run it from the Scripts menu and press **UP** during the countdown to open the settings. Go to **Tools & Scripts** and enable **MiSTer Monitor**, then save and let the update finish. (Requires Update All 2.10 or newer.)
- **MiSTer Companion** — open **Install Center**, find **MiSTer Monitor** and install it.

IMPORTANT Then run MiSTer_Monitor once from the Scripts menu. Downloading the files is not enough: this run enables auto-start on boot, sets log_file_entry=1 in **MiSTer.ini** — which is what lets the server see the loaded game, so without it no artwork ever appears — and starts the server. It is safe to run again at any time, and doing so after an update restarts the server on the new version.

- Insert the card in your display and power it.
- Load any game. Within a few seconds the display shows its artwork.

TIP While you are in the Update All settings, look under **Extra Content → Game Artwork DBs.** The artwork packs put images for 39 systems straight onto the MiSTer, in three packs that install separately: **Box Art**, **Screenshots** and **Title Screens.** Covered games appear instantly, without a ScreenScraper lookup. They are optional — everything works without them, and ScreenScraper still fills in whatever a pack does not cover. All three together take from about 6 GB to a little over 10 GB on the MiSTer's card, depending on how it is formatted; installation.md has the figures.

NOTE Running the **Downloader** on its own, without Update All? A drop-in database file does the same job — see **installation.md**. Use one route or the other, not both.

IMPORTANT **Console Mode** (Taki Udon's frontend) replaces **MiSTer.ini** with its own copy, which turns game detection back off. If you install or repair it, run **MiSTer_Monitor** once more afterwards and everything returns to normal.

To pause the monitor, run **MiSTer_Monitor_uninstall** from the Scripts menu; run **MiSTer_Monitor** again to re-enable it.

5 RetroAchievements (optional)

Your progress and live unlock popups on the display

This part lives **on the MiSTer**, not on the display's card. The first time the server runs it drops a ready-to-edit file in place — you only fill in two lines:

on the MiSTer — created automatically on first start

/media/fat/Scripts/.config/mister_monitor/ra_credentials.ini

```
[retroachievements]
username=YourRAUsername
api_key=YourWebAPIKey
```

Your **Web API key** is at retroachievements.org → Settings → Applications. Load a game with a published set and the RetroAchievements page comes alive: progress bar, points, and the full trophy list — tap any row for its description.

Credentials only stock MiSTer

The full page in **view-only** mode: the game's set, your cloud progress and every trophy — but play is not recorded.

odelot/Main_MiSTer fork RA_ cores

Achievements are **earned while you play**; a READY banner confirms it on each matched game, and unlocks pop within seconds.

Fork + debug=1 retroachievements.cfg

Popups become **instant** (with title and description), and CD systems — PlayStation, Saturn, Mega CD — resolve too.

IMPORTANT Each tier is optional and falls back gracefully to the one below. Without the credentials file, everything else in this guide works exactly the same.

Earning achievements on the MiSTer

Recording achievements needs odelot's RetroAchievements build of the MiSTer main binary, plus his modified core for each system. Two maintained routes install it for you, both keeping your stock cores intact:

- **Update All** — the RetroAchievements Cores database, maintained by odelot himself, is now built in. Run **Update All**, enable it in the settings screen, and it walks you through the retroachievements.cfg setup and the MiSTer.ini entries it needs.
- **MiSTer Companion** — install the **RetroAchievements Cores** entry from its **Install Center**. (mistercompanion.org)

Either route leaves a retroachievements.cfg on the MiSTer holding your RA **username and password**. That file belongs to odelot's build and is separate from the ra_credentials.ini above, which is this project's and takes an API key instead — you will end up with both, and that is expected.

```
two different files, both needed for the full experience
/media/fat/retroachievements.cfg     odelot's build  (user + password)
/media/fat/Scripts/.config/mister_monitor/ra_credentials.ini
                                     this project    (user + API key)
```

IMPORTANT Once the RA cores are installed an **RA_Cores** entry appears in the MiSTer OSD. Achievements are only active for cores launched from there — opening the same system from its usual menu entry runs the stock core and records nothing.

TIP Add debug=1 to retroachievements.cfg for sub-second unlock popups. Achievements only work on the systems odelot has published cores for — his repository lists the current set.

6 The web interface

Edit settings and manage the display's card from any browser

Once the display is on your network it runs a small web interface of its own. Point a browser at **the display's own IP address** on port 8080 — not the MiSTer's. The display shows its address on the boot screen once it connects, and on the **Network** page as **Monitor IP**, right below the MiSTer's.

```
edit config.ini
http://<display-ip>:8080/config
```

The page opens the card's **config.ini** in a text editor, exactly as it is written, comments and all. Saving writes it back to the microSD and keeps the previous version as **config.ini.bak** on the same card. Settings are read once at startup, so the page offers a **Reboot** button to apply them — the display restarts in a couple of seconds.

the settings page

http://<display-ip>:8080/files/download?path=/settings.html

The same file in a graphical editor, contributed by Rick van Soest: switches, sliders, drop-downs and drag-to-reorder lists, grouped in tabs, with the notes from config.ini under each field. The other pages link to it as **Settings page**. It rewrites only the values you change — comments and order stay as they are — keeps **config.ini.bak** like the text editor, and can reboot the display for you. The page is a file on the card, **settings.html**, copied in step 2; on a card prepared before 2.12.0, upload it to the root with the file browser below.

browse the microSD card

http://<display-ip>:8080/files

The same interface lists the card's contents: open folders, download a file to your computer, upload new ones, create folders, and delete what you no longer need. The most common use is artwork — delete the image for one game and it will be fetched again next time you load it, or upload your own to replace it.

NOTE Artwork you upload must be **baseline** JPEG. Progressive JPEGs are rejected with a message, because the display's decoder cannot read them. Most image editors offer the choice when exporting. Deleting covers files and empty folders; to remove a folder with contents, empty it first.

The interface has no password by default, which is fine on a home network. To lock it, set a password in the **[ui]** section of config.ini — the browser will then ask for it, with the username **admin**. Setting **web_config=false** disables the whole web interface.

config.ini — optional

```
[ui]
web_config=true
web_password=
```

The 2.8" CYD boards, the Tab5 and the Guition also serve a screenshot of what is on screen at **http://<display-ip>:8080/**. The 3.5" boards and the Waveshare 2.8" do not: their panels cannot be read back.

OK

You're done — a quick check

Working as intended looks like this

- Boot: the display connects to WiFi and finds the MiSTer by itself.
- Load a game: box art appears within seconds; the Game Info panel fills in year, developer and synopsis.
- RetroAchievements page: progress and trophies for the running game (view-only unless you installed the fork and open a RA enabled core).
- System pages: live CPU, memory, storage and network stats.
- Switch the MiSTer off: after three minutes the display dims to its standby screen, and it wakes by itself when the MiSTer comes back.
- Web interface: **http://<display-ip>:8080/config** opens the settings editor from any browser on your network.

↑

Updates over WiFi

From 2.12.0 on, new versions reach the display through the MiSTer — no cable

Every release now ships the display firmware next to the server, through the same updater you chose in step 4, so the MiSTer always holds the version that matches it. Updating the display is then a tap:

- Run your updater on the MiSTer, then **MiSTer_Monitor** once from the Scripts menu, which restarts the server on the new version.
- The display notices the newer version and shows a green banner for a few seconds: **UPDATE TO x.y.z — TAP TO INSTALL**. Tap it. The display shows the progress and restarts by itself; nothing on the microSD changes.
- Missed the banner? The version panel on the main screen stays orange. Tap the main screen to bring the banner back, then tap the banner.
- From a browser, **http://<display-ip>:8080/update** has an install button too, and can install a .bin file from your computer as well.
- To install without a tap, set ota_auto=true in the **[update]** section of config.ini.

It is safe to do away from a cable. A download that stops halfway or arrives damaged is discarded before it can start, and the display keeps running the version it had.

IMPORTANT **Coming from 2.11.1 or earlier?** Those versions cannot update themselves. Flash 2.12.0 once with the web flasher, as in step 1 — your card and settings stay as they are — and from then on updates arrive over WiFi. On the Cheap Yellow Displays this flash also moves the board to the two-slot memory layout that updates need; if the **/update** page ever reports a **single app slot**, flash once more with the web flasher.

NOTE An update replaces the firmware and nothing else. Options added to config.ini keep their defaults until you set them — the settings page (step 6) lists them all the same, and adds them to your file when you change them. New files for the card, such as settings.html, are uploaded from **/files**. **Guition:** replace c6_firmware.bin in the root of the card with the one from the new release as well.

.ini

Everything config.ini can do

Every option, grouped by what you want to achieve — all of it optional

Steps 2 and 3 filled in the only values the display really needs. Everything else in **config.ini** already has a sensible default, and this section is the map of what you can change. The file itself carries a comment above every key, and **configuration.md** goes into more detail.

- Edit it from a browser, as text at **http://<display-ip>:8080/config** or with the settings page (step 6). Settings are read once at startup, so press **Reboot** on either page to apply a change.
- A missing key keeps its default, so an older config.ini keeps working after an update. Newer options are in the config.ini under **SD_card_content/** for your board — copy their lines across.
- Section names such as **[ui]** only organise the file; keys are found by name.

key_name	default

How to read the rows below: the name of the option in config.ini, its default in the field on the right, and what it does.

Network and accounts

ssid	—
password	—

Your WiFi network, without quotes.

ip	blank

Blank: the display finds the MiSTer by itself. Set: it follows that MiSTer and no other — see step 2.

ss_user	—
ss_pass	—

Your free ScreenScraper member account.

region	wor

Artwork region tried first: **wor**, **us**, **eu** or **jp**. The others follow in a fixed order.

timeout	30000
retries	2
use_https	false

ScreenScraper connection: request timeout in ms, retries when the service is overloaded, and HTTPS — enable that one only if plain HTTP stops working.

Which artwork is downloaded

Four lists decide which image types are shown, each tried left to right until one exists. For games, a box, screenshot or title screen in the artwork packs on the MiSTer (step 4) is taken from there; everything else is requested from ScreenScraper. Reorder the lists to taste, or remove a type to skip it.

game_media_order

Console and computer games. Box art first.

arcade_media_order

Arcade games. Fan art and marquees first — most have no box.

core_media_order

The system itself (Mega Drive, Amiga…). Logo wheels first.

arcade_subsystem_media_order

Arcade platforms (CPS1, Sega Classics…). Logo wheels first.

the image types you can list

for games	box3d box2d fanart marquee screenshot titlescreen mix1 mix2
for systems	screenmarquee background photo illustration
for both	wheel-steel wheel-carbon wheel

auto_download `true`

Fetch missing artwork from ScreenScraper.

force_core_redownload `false`

force_game_redownload `false`

Fetch again even when the image is already on the card. Set for one boot, then back to **false**.

max_image_size `500000`

download_timeout `30000`

Largest image accepted, in bytes, and the download timeout in ms.

base_path `/cores`

default_image `/cores/menu.jpg`

alphabetical_folders `true`

Where images live on the card, the default image, and whether they are sorted into lettered folders. Rarely worth changing.

What the image screen shows

image_mode `rotate`

rotate alternates the game's artwork and the system's; **game** and **system** stay on one and never switch; **cycle** stays on the game and alternates its screenshot, title screen and box. Cycle takes them from the artwork packs on the MiSTer, so it needs the Screenshots and Title Screens packs (step 4).

core_image_timeout `30000`

system_image_timeout `0`

How long each slide stays up, in ms. The second gives the system slide its own time; 0 means the same as the first.

info_in_rotation `false`

Adds the GAME INFO panel to the rotation as a third slide.

image_upscale `false`

image_upscale_max `2.5`

Grow artwork that is smaller than the display area until it fills it, up to that multiplier. Most visible on the Tab5 and the Guition.

screenshot_scaling `fill`

How screenshots and title screens fill the image area. **fill** always fills it, with crisp pixels and a slight softening on some edges; **integer** grows them only by whole multiples, every pixel square, so some stay smaller. Box art is not affected.

kiosk_mode `false`

kiosk_hide_delay_ms `5000`

Give the whole panel to the artwork and hide the footer. A tap brings the footer up for that many ms; a second tap works as usual.

art_inset_left `0`

Waveshare 2.8" only. Pixels kept free on the left edge, 0–120, for cases whose window is not centred on the panel; the artwork is centred in the rest.

NOTE Kiosk mode asks ScreenScraper for taller images, so artwork already on the card shows a black band above and below until it is fetched again. Set both force_…_redownload keys to **true** for one boot, or delete the images from the web interface.

The Game Info panel

info_lang `en`

Language of the synopsis and genre: **en es pt fr de it**. Falls back to English.

info_scroll_auto `true`

info_scroll_step_ms `2000`

Scroll the synopsis by itself, and how fast, in ms per line — higher is slower.

info_synopsis_max `2000`

Longest synopsis kept per game, in characters. A memory limit; rarely needs changing.

Standby

With the MiSTer switched off there is nothing to show, so the display dims to a standby screen; it can do the same while the MiSTer sits in its menu. It wakes by itself when the MiSTer answers again or the core or game changes, and at a touch.

standby_screen `clock`

clock: a large clock, the date — with the outside temperature after it, once a place is set below — whether the MiSTer is online and what it is running. **minimal**: the MiSTer logo and a small state dot. The clock needs a **timezone**; until it has one, minimal is shown.

standby_when_offline `true`

standby_offline_min `3`

Go to standby once the MiSTer has been unreachable for that many minutes. **false** turns it off.

standby_idle_min `0`

Also go to standby after this many minutes with no core change, game change, achievement or touch — even with the MiSTer running. 0 = never.

standby_in_menu `false`

standby_menu_delay `0`

Also go to standby while the MiSTer is switched on but sitting in its menu, with no core loaded, after that many seconds there — 0 = at once. Loading a core wakes the display; a tap wakes it for half a minute.

standby_brightness `10`

Backlight level in standby, 0-255. 0 switches it off.

standby_dim 100

Darkens what the standby screen draws, as a percentage. For Tab5 units whose backlight only switches on and off: try 30. Leave at 100 elsewhere.

The clock and the outside temperature

timezone blank

Where you are — a name from the list below, or a POSIX TZ string. Blank: no clock anywhere.

ntp_server pool.ntp.org

clock_24h true

The time server, and 24-hour or AM/PM time.

weather_city blank

Your town, for the outside temperature on the clock screen. It comes from Open-Meteo — free, no account or key — and nothing is fetched until a place is set.

weather_lat blank

weather_lon blank

Coordinates instead of a name. They win when both are set.

weather_unit c

c or **f**.

timezone names — each covers every country that shares its time

uk portugal iceland spain nigeria greece southafrica turkey

dubai india thailand china japan sydney newzealand

argentina venezuela colombia mexico

useastern uscentral usmountain uspacific

```
[ui]
timezone=spain
```

TIP **spain** also covers France, Germany, Italy, the Netherlands, Poland and Sweden; **uk** covers Ireland; **japan** covers South Korea. The full list of countries is in config.ini, above the key. Daylight saving changes are handled.

Screen, sound, web interface and updates

flip_display false

Turn the picture 180°, touch included, for cases that hold the board upside down.

touch_invert_x false

touch_invert_y false

CYD 3.5" only. Mirror one touch axis, for boards whose touch layer reports it reversed: a tap near one edge acts on the opposite one. The LCDwiki E32R35T needs **touch_invert_y=true**.

scroll_speed_ms 300

scroll_pause_start_ms 2000

scroll_pause_end_ms 3000

Scrolling of long names in the footer: ms per step — lower is faster — and the pauses before it starts and before it resets.

sound true

sound_volume 128

Tab5 only. Silence the boot jingle and touch beeps, or set the volume, 0–255.

panel_rev v1

Guition only. v2 for units from batch 2624 onward, and every -V3 unit — see step 2.

web_config true

web_password blank

Switch the web interface off, or protect it with a password (username **admin**) — see step 6.

ota_auto false

Install an offered firmware update by itself, without a tap — see Updates over WiFi.

debug false

Verbose messages on the serial port, for troubleshooting.

The Waveshare 2.8" desk clock

Waveshare 2.8" only. A desk clock for the time the MiSTer sits in its menu: the time, the year and a smiling face in turn, drawn as dots on a grey LCD-style background, with the outside temperature, the weekday and the date. It needs a **timezone**.

idle_clock — false

idle_clock_seconds — 30

Turn it on, and how long each of the three screens stays up, in seconds.

idle_lcd_left — 0

A black border on the left, 0–120 px, for cases whose window is not centred on the panel.

idle_day_names — SUN,…,SAT

idle_date_order — md

The seven weekday names, Sunday first, and the date as month.day (**md**) or day.month (**dm**).

idle_face_file — blank

A face of your own, as a text file on the card — configuration.md explains the format. Blank: the built-in one.

links

Project	github.com/chipster6502/MiSTer_monitor
Docs	installation.md configuration.md
Flasher	chipster6502.github.io/MiSTer_monitor/flasher/
RetroAchiev.	retroachievements.org
RA fork	github.com/odelot/Main_MiSTer
Update All	github.com/theypsilon/Update_All_MiSTer
Companion	mistercompanion.org
Artwork	github.com/chipster6502/MiSTer_artwork_pack
Display UI	http://<display-ip>:8080/config
Updates	http://<display-ip>:8080/update
Server page	http://<mister-ip>:8081/

MiSTer Monitor is free software — firmware under GPL-3.0, server under AGPL-3.0. Found a rough edge in this guide? Open an issue; it helps the next person.